CateDocs: A New Visual Analytics Approach to Exploring High Dimensional Categorical Datasets

Yueqi Hu, UNC Charlotte
Xiaoke Huang, Kent State University
Chong Zhang, UNC Charlotte
Yingyu Wu, Kent State University
Jing Yang, UNC Charlotte
Ye Zhao, Kent State University
Scott Barlowe, Western Carolina University
Wei Chen, Zhejiang University

ABSTRACT

Most existing multidimensional visualization techniques do not work well for high dimensional categorical datasets. The major challenges include preserving the discrete nature of the data and visually exploring the high dimensional space. In this poster, we propose a new visual analytics approach for high dimensional categorical data. Our methodology is to convert a categorical dataset into a document corpus and then apply advanced document analysis and visualization techniques to the corpus. Two prominent knowledge discovery tasks, namely cluster analysis and multivariate analysis, are supported. For cluster analysis, the Latent Dirichlet Allocation (LDA) topic model is employed to discover subspace clusters in a categorical dataset. The clusters are then visualized in a semantically rich visualization for interactive visual analysis. For multivariate analysis, LDA is used for dimension reduction and optimal rule mining is used to discover rules describing multivariate relationships in the reduced subspaces. The effectiveness of this approach has been illustrated by case studies on real datasets.

Keywords: Categorical data visualization, high dimensional data

1 INTRODUCTION

Categorical datasets with large numbers of dimensions and categories are common in real applications. It is important to explore these datasets for decision making. Unfortunately, most existing visualization approaches do not scale well to large categorical datasets. Many challenges need to be addressed, such as effectively guiding users to navigate in the high dimensional space and supporting cluster analysis, prediction, and other investigative tasks.

We propose a novel approach to visual analytics of high dimensional categorical datasets. Our strategy is to regard category values as a vocabulary and then each data item becomes a document built upon this vocabulary. This allows a categorical dataset to be studied as a corpus. Advanced document analysis and visualization techniques can then be used for visual analytics of categorical datasets. They excel in handling high dimensional keyword spaces, processing large numbers of documents, dealing with missing values, and presenting semantics, all of which are desired in high dimensional categorical data analysis. We call this methodology **CateDoc**. In this poster, we present two implementation examples of CateDoc: subspace cluster visual analytics and risk/preventive rule mining.

A novel subspace cluster visual analytics approach is proposed, which is named **TopicCards** (Figure 1). It employs Latent Dirichlet Allocation (LDA) [1] for subspace cluster analysis. LDA can automatically identify the topic structure from an unlabeled document corpus. After applying LDA on the corpus converted from a categorical dataset, a subspace cluster is constructed from each topic identified by LDA. The subspace is defined by category values highly related to the topic and data items highly related to the topic are members of the cluster.

Risk/preventive rule mining helps analysts understand the relationship between a dependent variable and many independent variables. We propose using LDA to find meaningful lower-dimensional subspaces for risk/preventive rule mining. Optimal Risk Rule Mining [4] was then used to discover rules from the reduced subspaces constructed according to LDA results. Results of an experiment showed that very few rules were missing while the time cost was reduced significantly using the LDA approach.

We employ LDA for subspace clustering and dimension reduction because it has the following merits: (1) LDA is able to reveal topics widely distributed in a corpus as well as salient topics in relatively small subsets of documents. This allows user to identify small but significant clusters; (2) LDA scales well to the number of dimensions and categories as well as the record size; (3) The resulting clusters / subspaces are described by a set of category values, which are very intuitive to interpret and visualize.

The major contributions include: (1) Our subspace cluster visual analysis approach yields good results on high-dimensional categorical datasets with missing values; (2) We propose a novel paradigm of exploring high dimensional categorical datasets by applying text document analysis and visualization techniques.

2 SUBSPACE CLUSTER VISUAL ANALYTICS

Data Analysis Each record in a dataset is converted to a "document" represented by a bag of category values. For example, the following is the "document" of a data item in the MovieLens dataset [3] (each data item in the dataset records the information of a movie and a user who rated the movie): *movie_name=Toy_Story year=90s genres=animation genres=children genres=comedy age=under18 gender=f.* Audience occupation, which is a missing value in this record, is skipped in this document but appears in other documents where its values are available.

The corpus formed by all documents is sent to the Standford Topic Modeling Toolbox (STMT) [5]. The output of STMT is two matrices. One is the topic-category value matrix presenting the distribution of the topics over the category values. The other is the topic-data item matrix recording the distribution of the data items over the topics. For each topic, the category values are sorted by their weights in that topic and the top values form the subspace containing a cluster. The weight of a data item on the topic indicates its association with the cluster. The data items with higher weights than a given threshold are considered a member of the cluster.

No category values are dropped in LDA. Both high frequency values and low frequency values can convey useful information in the clusters. Since high frequency category values can gain high weights in many topics, we remove a category value if its local support (the ratio of data items with the keyword in the cluster) is lower than its global support (the ratio of data items with the category value in the dataset). Determining a proper topic number for LDA is remained for future work. Alternatively, LDA is executed several times with different specified numbers of topics, which allows

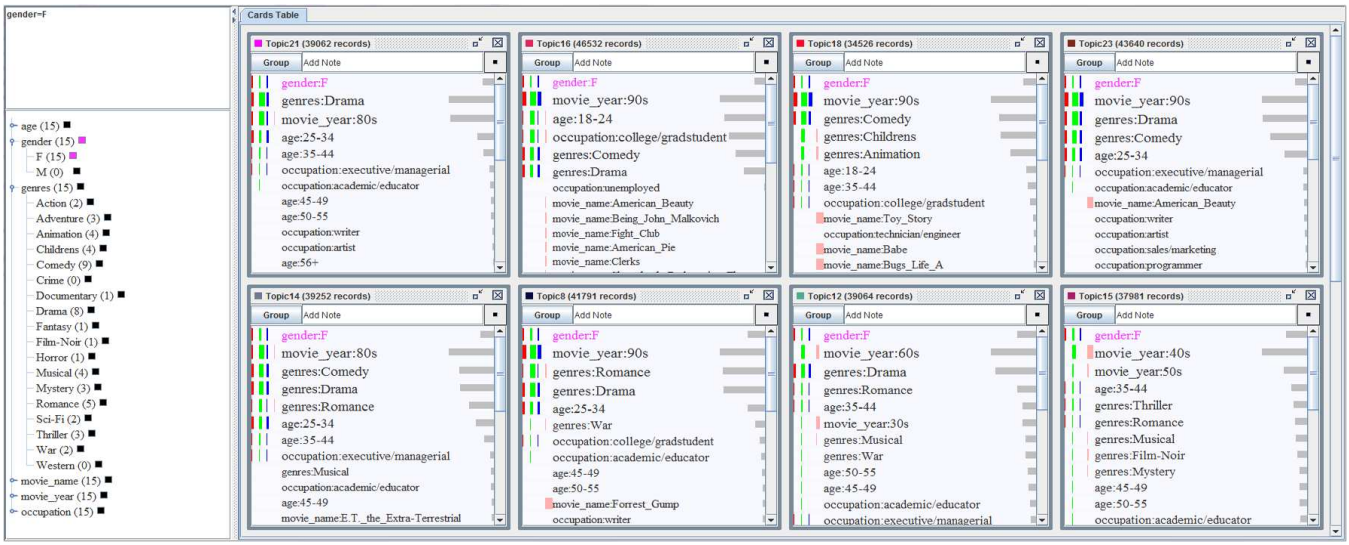


Figure 1: Clusters containing category value gender=F in the MovieLens dateset. The category values in the clusters demonstrate the subtle difference among the clusters.

users to examine either coarse or fine clusters by simply loading the results of model runs with different numbers of topics.

Visualization Each subspace cluster is represented by a **Topic Card**. The cards can be either browsed in an overview (see Figure 1) or examined individually. Each card displays the category values defining the subspace in a list. The colorful blocks on the left of a category value display statistics of the value, such as the percentage of data items in the cluster containing the value. The grey bar on the right reveals the weight of the value in the topic, which is also displayed by the font size of the text. Users can identify interesting clusters and category values according to the text, blocks and bars.

The concentration of the weight distribution of data items on a topic is used to measure the interestingness of the cluster. The clusters are ordered by interestingness in the overview to provide an informative initial view. A search / coloring interface is provided on the left of the overview, where category values are organized as leaf nodes of the dimensions. It establishes a connection between the clusters and the dimensions, since users can click dimensions / categorical values on it to select clusters and highlight interesting dimensions in the cards. Two case studies on real world high dimensional datasets have been conducted. A video demo presenting these case studies can be found from **http://www.youtube.com/watch?v=W2Kt7WKIMTI**

3 Risk/Preventive Rule Mining

Optimal Risk Rule Mining (ORRM) [4], which identifies risk/preventive patterns, becomes slow when the dimensionality is high. To overcome this bottleneck, we use LDA for dimension reduction and then apply ORRM on the reduced subspace. This approach has been evaluated through an experiment.

Two approaches, namely ORRM-only and LDA-ORRM, were compared in the experiment. The ORRM-only approach directly applied ORRM to the Mushroom dataset (8,214 data items, 22 independent variables, and 1 dependent variable) [2]. The LDA-ORRM approach consists of five steps: (1) converting data items to documents; (2) applying LDA to the corpus with M topics specified; (3) identifying topics containing a value of interest in the dependent variable and defining a subspace using category values highly weighted on those topics; (4) projecting the dataset to the subspace and applying ORRM to it.

The ORRM-only approach was conducted once. There were multiple runs of the LDA-ORRM approach with different M values. Set A is defined as the distinct categories values in the rules generated by a LDA-ORRM run. Set B is defined as the distinct categories values in the rules generated by the ORRM-only approach. The running time of a LDA-ORRM approach is the total running time of its four steps. Three measures were calculated: (1) Time ratio = LDA-ORRM running time / ORRM-only running time; (2) Precision = Size of (Set A ∩ Set B) / Size of Set B; (3) Recall = Size of (Set A ∩ Set B) / Size of A. The results are reported in Figure 2. It suggests that LDA-ORRM was much faster than ORRM-only with a small amount of information loss.

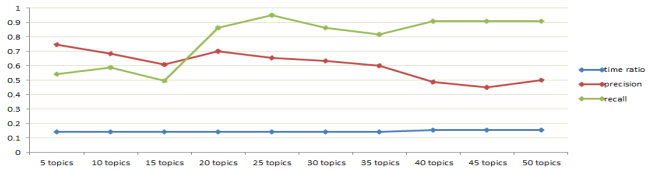


Figure 2: The results of the experiment comparing ORRM-only and LDA-ORRM approaches.

4 Conclusion

In this poster, we present CateDocs, a methodology that employs document analysis and visualization techniques for visual analytics of categorical data. The effectiveness of CateDocs is demonstrated using two important knowledge discovery tasks, namely cluster analysis and risk/preventive pattern analysis.

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